

Subject Description Form

Subject Code	SFT5107
Subject Title	Recycling via Technology and Design
Credit Value	3 credits
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	This subject has the following objectives: 1. To explore the concepts and principles of recycling with technology and product design (RTPD). 2. To analytically assess the performance of the alternative options of RTPD. 3. To explore the latest industrial advance and development in RTPD. 4. To apply different tools for exploring RTPD related problems.
Intended Learning Outcomes	Students who have successfully completed the subject shall be able to: a. Apply scientific tools and methods to address RTPD problems. b. Understand the latest trend in RTPD. c. Propose appropriate scientifically sound solutions for solving the problems in RTPD. d. Present and discuss in English effectively and clearly. e. Co-operate and contribute as a team member/leader.
Subject Synopsis/ Indicative Syllabus	(I) Concepts of recycling with technology and product design (RTPD) in apparel and textiles The critical concepts and principles in RTPD (5R concept). Technological advances in RTPD. How to evaluate different configurations of RTPD. Sustainable product development and recycling. “5R” full words are necessary. (II) Implementation of 5R concept Reviews of the classical production and product development system to implement the 5R concept. Technological and design oriented principles. Real case studies. (III) Use of technology The importance of material science technology and information technology in RTPD. Roles of nanotechnology, sustainable fabrics, organic fibers, etc. The importance of information systems such as IoT, RFID and blockchain for enhancing provenance of material flows along the supply chain system. “all abbreviations need to be defined firstly, allowing young students to fully understand” (IV) Design for recycling From clothing and apparel design perspective, examine the principles which are critical to achieving sustainable product design for recycling. How artistic design and functional design can strike a balance. (V) Contemporary Issues in RTPD A discussion on the latest developments and hot topics in RTPD in the apparel and textiles industry. Real case based analyses.

Teaching/ Learning Methodology	This subject comprises lectures and tutorials. Students are required to examine contemporary issues in the areas of RTPD, and have group discussions. Guided readings will be suggested for private study to complement the subject							
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					
			a	b	c	d	e	
	1. Short Assignments/Tutorials	25%	✓	✓	✓			
	2. Group projects	25%		✓	✓	✓	✓	
	3. Final exam	50%	✓	✓	✓			
	Total	100%						
	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Short assignments (including home-works, class-works, and tutorial exercises), both individual and group-based, and tests are used to assess the first three learning outcomes related to the professional knowledge on EFPSF. A comprehensive group project, which includes case analysis and presentation, is employed to assess four learning outcomes, including the two generic outcomes d and e.							
Student Study Effort Expected	Class Contact:							
	▪ Lectures		26 hrs. (i.e.; 13*2 hrs)					
	▪ Tutorials		13 hrs. (i.e.; 13*1 hr)					
	Other Study Effort:							
	▪ Assignments + Reading		70 hrs.					
	Total Student Study Effort		109 hrs.					

Reading List and References	Books: Blackburn, R. (Editor) Sustainable Apparel: Production, Processing and Recycling (Woodhead Publishing Series in Textiles) 1st Edition, 2015. Journal papers: Choi, T.M., Y.J. Cai, B. Shen. Sustainable fashion supply chain management: a system of systems analysis. IEEE Transactions on Engineering Management, 66(4), 730-745, 2019. Choi, T.M., Y.J. Cai. Impacts of lead time reduction on fabric sourcing in apparel production with yield and environmental considerations. Annals of Operations Research, published online, 2018. Choi, T.M., W.K. Yeung, T.C.E. Cheng, X. Yue. Optimal scheduling and the value of RFID technology in garment manufacturing systems. IEEE Transactions on Engineering Management, 65(1), 72-84, 2018. Fung, Y.N., T.M. Choi, R. Liu. Sustainable planning strategies in supply chain systems: Proposal and applications with a real case study in fashion. Production Planning and Control, in press, 2019. Guo, S., T.M. Choi, B. Shen. Green product development under competition: a study of the fashion apparel industry. European Journal of Operational Research, 280(2), 523-538, 2020. Ho, H.P.Y., T.M. Choi. A five-R analysis for sustainable fashion supply chain management in Hong Kong: a case analysis. Journal of Fashion Marketing and Management, 16, 161-175, 2012.
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